

Geodesic Motion in Kerr Spacetime — Summary

Geodesic Equation and Reparametrization

A free-falling particle follows a geodesic, obtained from the Euler–Lagrange equations of $\mathcal{L} = \frac{1}{2}g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu$:

$$\frac{d^2x^\mu}{d\tau^2} + \Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} = 0. \quad (1)$$

The normalization $g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu = \kappa$ ($\kappa = -1$ timelike, $\kappa = 0$ null) is an exact first integral and serves as a numerical diagnostic. For astrophysical applications, reparametrizing by coordinate time t yields the more practical form:

$$\frac{dv^i}{dt} = -\Gamma_{\alpha\beta}^i v^\alpha v^\beta + \Gamma_{\alpha\beta}^t v^\alpha v^\beta v^i, \quad (2)$$

where $v^i = dx^i/dt$ and $v^t = 1$. The second term corrects for t not being an affine parameter and must not be omitted.

The Kerr Metric: BL vs. KS

In **Boyer–Lindquist (BL)** coordinates the Kerr metric is diagonal in r and θ but singular at $\Delta = r^2 - 2Mr + a^2 = 0$ (horizons) and at $\sin\theta = 0$ (poles). These coordinate singularities make BL ill-suited for numerical integration near the horizon or the symmetry axis.

Kerr–Schild (KS) coordinates (t, x, y, z) remove both coordinate singularities. The metric takes the linear form $g_{\mu\nu} = \eta_{\mu\nu} + f l_\mu l_\nu$, with inverse $g^{\mu\nu} = \eta^{\mu\nu} - f l^\mu l^\nu$, where:

$$f = \frac{2Mr^3}{r^4 + a^2 z^2}, \quad l_\mu dx^\mu = dt + \frac{rx + ay}{r^2 + a^2} dx + \frac{ry - ax}{r^2 + a^2} dy + \frac{z}{r} dz. \quad (3)$$

The BL radius r is defined implicitly by $(x^2 + y^2)/(r^2 + a^2) + z^2/r^2 = 1$, solved explicitly as $r^2 = K + \sqrt{K^2 + a^2 z^2}$ with $K = (x^2 + y^2 + z^2 - a^2)/2$. The only remaining singularity is the physical ring singularity at $r = 0, z = 0$.

Derivative Ladder and Geodesic Acceleration

All metric derivatives $\partial_k g_{\mu\nu} = (\partial_k f) l_\mu l_\nu + f(\partial_k l_\mu) l_\nu + f l_\mu (\partial_k l_\nu)$ are computed layer by layer, each building on the previous one.

Layer 1 — Derivatives of r . From $r^2 = K + W$ with $K = (x^2 + y^2 + z^2 - a^2)/2$ and $W = \sqrt{K^2 + a^2 z^2}$:

$$\partial_x K = x, \quad \partial_y K = y, \quad \partial_z K = z, \quad \partial_x W = \frac{Kx}{W}, \quad \partial_y W = \frac{Ky}{W}, \quad \partial_z W = \frac{z(K + a^2)}{W}. \quad (4)$$

Combining via $r^2 = K + W$ and $\rho^2 \equiv r^2 + a^2$:

$$\partial_x r = \frac{xr}{2W}, \quad \partial_y r = \frac{yr}{2W}, \quad \partial_z r = \frac{z\rho^2}{2rW}. \quad (5)$$

Layer 2 — Derivatives of $\mathcal{S}$. From $\mathcal{S} = r^4 + a^2 z^2 = 2r^2 W$:

$$\partial_x \mathcal{S} = \frac{2xr^4}{W}, \quad \partial_y \mathcal{S} = \frac{2yr^4}{W}, \quad \partial_z \mathcal{S} = \frac{2r^2 \rho^2 z}{W} + 2a^2 z. \quad (6)$$

Layer 3 — Derivatives of f . Using $f = Mr/W$ (which follows from $f = 2Mr^3/\mathcal{S}$ and $\mathcal{S} = 2r^2 W$):

$$\partial_i f = \frac{M}{W} \left(\partial_i r - \frac{r}{W} \partial_i W \right). \quad (7)$$

Layer 4 — Derivatives of l_μ . The spatial components $l_x = (rx + ay)/\rho^2$, $l_y = (ry - ax)/\rho^2$, $l_z = z/r$ give:

$$\partial_i l_x = \frac{r \delta_{ix} + a \delta_{iy}}{\rho^2} + \mathcal{L}_x \partial_i r, \quad (8)$$

$$\partial_i l_y = \frac{r \delta_{iy} - a \delta_{ix}}{\rho^2} + \mathcal{L}_y \partial_i r, \quad (9)$$

$$\partial_i l_z = \frac{\delta_{iz}}{r} + \mathcal{L}_z \partial_i r, \quad (10)$$

where the shape factors, fully determined at Layer 1, are:

$$\mathcal{L}_x = \frac{x(a^2 - r^2) - 2ary}{\rho^4}, \quad \mathcal{L}_y = \frac{y(a^2 - r^2) + 2arx}{\rho^4}, \quad \mathcal{L}_z = -\frac{z}{r^2}. \quad (11)$$

Layer 5 — Metric derivatives. Since the metric is stationary ($\partial_t g_{\mu\nu} = 0$), only spatial derivatives are needed:

$$\partial_k g_{\mu\nu} = (\partial_k f) l_\mu l_\nu + f (\partial_k l_\mu) l_\nu + f l_\mu (\partial_k l_\nu). \quad (12)$$

Geodesic acceleration. The Christoffel symbols split as $\Gamma_{\alpha\beta}^\mu = \bar{\Gamma}_{\alpha\beta}^\mu - \frac{f}{2} l^\mu \mathcal{G}_{\alpha\beta}$, where $\bar{\Gamma}$ uses $\eta^{\mu\nu}$ in place of $g^{\mu\nu}$ and $\mathcal{G}_{\alpha\beta} = l^\sigma (\partial_\alpha g_{\beta\sigma} + \partial_\beta g_{\alpha\sigma} - \partial_\sigma g_{\alpha\beta})$. Substituting into (2):

$$\frac{dv^i}{dt} = \underbrace{-\bar{\Gamma}_{\alpha\beta}^i v^\alpha v^\beta + \bar{\Gamma}_{\alpha\beta}^i v^\alpha v^\beta v^i}_{\bar{F}^i} + \frac{f}{2} \mathcal{G} (l_i + v^i), \quad (13)$$

where $\mathcal{G} = \mathcal{G}_{\alpha\beta} v^\alpha v^\beta$ is a scalar computed once per step. The full evaluation order at each ODE call is therefore: Layer 1 $\rightarrow$ 2 $\rightarrow$ 3 $\rightarrow$ 4 $\rightarrow$ 5 $\rightarrow$ assemble $\bar{\Gamma}$, $\mathcal{G}$, F^i .

Constants of Motion and Integrability

The two Killing vectors of the Kerr metric yield the conserved specific energy $E = -g_{t\mu} u^\mu$ and axial angular momentum $L_z = g_{\phi\mu} u^\mu$. A third constant, the **Carter constant**:

$$\mathcal{Q} = p_\theta^2 + \cos^2 \theta \left(a^2 (\mu^2 - E^2) + \frac{L_z^2}{\sin^2 \theta} \right), \quad (14)$$

arises from a rank-2 Killing tensor (beware: this is the form in the BL coordinates!). Together with the normalization constraint, these four constants render the Kerr geodesic problem *completely integrable*. In KS coordinates $\mathcal{Q}$ does not assume a simple form. It is evaluated via its Killing-tensor definition $\mathcal{Q} = K^{\mu\nu} u_\mu u_\nu - (L_z - aE)^2$, where you can prove

$$K^{\mu\nu} = \Sigma (k^\mu n^\nu + k^\nu n^\mu) + r^2 g^{\mu\nu}, \quad (15)$$

with

$$k^\mu = \left(\frac{r^2 + 2Mr + a^2}{\Delta}, \frac{xr + ya}{r^2 + a^2} + \frac{2ay}{\Delta}, \frac{yr - xa}{r^2 + a^2}, \frac{2axz}{\Delta r} \right), \quad (16)$$

$$n^\mu = \left(\frac{\Delta}{2\Sigma}, -\frac{\Delta}{2\Sigma} \frac{xr + ya}{r^2 + a^2}, -\frac{\Delta}{2\Sigma} \frac{yr - xa}{r^2 + a^2}, -\frac{\Delta}{2\Sigma} \frac{z}{r} \right). \quad (17)$$

Coordinate Transformation BL $\leftrightarrow$ KS

The KS spatial coordinates follow from $(r + ia)e^{i\phi_{\text{BL}}} \sin \theta = x + iy$, $z = r \cos \theta$. The KS time differs from BL time: $dt_{\text{KS}} = dt_{\text{BL}} + (2Mr/\Delta) dr$, which is regular at the horizon. The full 4×4 Jacobian of the transformation takes the block form:

$$\Lambda^\mu{}_\nu = \begin{pmatrix} 1 & 2Mr/\Delta & 0 & 0 \\ 0 & & J^i{}_j & \end{pmatrix}, \quad (18)$$

where $J^i_j = \partial x^i_{\text{KS}} / \partial x^j_{\text{BL}}$ is the 3×3 spatial Jacobian. Since the four-velocity transforms covariantly as $u^\mu_{\text{KS}} = \Lambda^\mu_\nu u^\nu_{\text{BL}}$, the coordinate velocity transformation follows immediately from the ratio u^i/u^0 :

$$v^i_{\text{KS}} = \frac{J^i_j v^j_{\text{BL}}}{1 + \frac{2Mr}{\Delta} v^r_{\text{BL}}}, \quad (19)$$

where the denominator $= u^0_{\text{KS}}/u^0_{\text{BL}}$ deviates from unity whenever $v^r_{\text{BL}} \neq 0$.

Numerical Accuracy and Diagnostics

The normalization constraint in KS form,

$$\mathcal{N} = -(v^t)^2 + \delta_{ij} v^i v^j + f(l_\mu v^\mu)^2 = \kappa/\dot{t}^2, \quad (20)$$

must be monitored at every step. A growing $|\mathcal{N}|$ signals excessive step size; a spike near the horizon in BL coordinates signals a coordinate singularity (switch to KS). The conserved energy E provides an independent check via the fractional drift $\delta E(t) = |E(t) - E(0)|/|E(0)|$.